

EX NO : 2

DATA CLEANING

DATE : 18-06-26

AIM:

Perform data cleaning operations on Employee and Dirty Sales datasets.

PROBLEM STATEMENT:

Q1: Using the Employee dataset provided

<https://www.kaggle.com/code/faressayah/ibm-hr-analytics-employee-attrition-performance/input>

Clean the IBM Employee dataset by handling missing values, removing duplicates, standardizing categorical values, correcting datatypes, removing salary outliers using IQR, and saving the cleaned dataset.

Q2: Dirty Cars dataset

https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training?utm_source=chatgpt.com

Clean the Dirty Sales dataset by handling missing values, removing duplicate transactions, removing invalid transactions, converting dates, standardizing text fields, and saving the cleaned dataset.

PART A – EMPLOYEE:

1. Identify missing values and handle them appropriately.
2. Remove duplicate records.
3. Standardize categorical values such as Gender and Department. (make them into throughout the dataset)
Correct incorrect data types (Examine the data type of each attribute using df.info()).
4. Identify columns whose data types do not match the nature of the data convert them to appropriate types.)
5. Detect and remove outliers in Salary using the IQR method.
6. Save the cleaned dataset as cleaned_employee.csv.
7. Compare the number of records before and after cleaning.
8. Summarize the cleaning operations performed.

CODE:

```
import pandas as pd

# 1. Load Dataset
df = pd.read_csv("D:\SEM_5\ML\datasets\Data cleaning\employee\WA_Fn-UseC_-HR-Employee-Attrition.csv")
print("Original Shape:", df.shape)
records_before = len(df)

# 2. Missing Values
print("\n===== MISSING VALUES =====")
print(df.isnull().sum())
missing_before = df.isnull().sum().sum()
for col in df.columns:
    if df[col].dtype == "object":
        df[col] = df[col].fillna(df[col].mode()[0])
    else:
        df[col] = df[col].fillna(df[col].median())
missing_after = df.isnull().sum().sum()

# 3. Remove Duplicate Records
duplicate_count = df.duplicated().sum()
df = df.drop_duplicates()
print("\nDuplicates Removed:", duplicate_count)
```

4. Standardize Categorical Values

```
if "Gender" in df.columns:
    df["Gender"] = (df["Gender"].astype(str).str.strip().str.lower().replace({"male": "Male", "female": "Female"}))
if "Department" in df.columns:
    df["Department"] = (df["Department"].astype(str).str.strip().str.title())
```

5. Check and Correct Data Types

```
print("\n===== DATA TYPES BEFORE =====")
df.info()
for col in df.columns:
    if ("Age" in col or "Income" in col or "Rate" in col or "Years" in col):
        df[col] = pd.to_numeric(df[col], errors="coerce")
print("\n===== DATA TYPES AFTER =====")
df.info()
```

6. Detect and Remove Salary Outliers Using IQR Method

```
salary_col = "MonthlyIncome"
Q1 = df[salary_col].quantile(0.25)
Q3 = df[salary_col].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - (1.5 * IQR)
upper_bound = Q3 + (1.5 * IQR)
outliers_removed = len(df [ (df[salary_col] < lower_bound) | (df[salary_col] > upper_bound) ])
df = df[ (df[salary_col] >= lower_bound) & (df[salary_col] <= upper_bound)]
```

7. Save Cleaned Dataset

```
df.to_csv("cleaned_employee.csv", index=False)
records_after = len(df)
```

8. Cleaning Summary

```
print("\n===== CLEANING SUMMARY =====")
print("Records Before Cleaning :", records_before)
print("Records After Cleaning :", records_after)
print("Missing Values Found :", missing_before)
print("Missing Values Removed :", missing_before - missing_after)
print("Duplicates Removed :", duplicate_count)
print("Salary Outliers Removed :", outliers_removed)
print("\nCleaned dataset saved as:")
print("cleaned_employee.csv")
```

OUTPUT:

1.

```
Original Shape: (1470, 35)
```

2.

```
===== MISSING VALUES =====  
Age 0  
Attrition 0  
BusinessTravel 0  
DailyRate 0  
Department 0  
DistanceFromHome 0  
Education 0  
EducationField 0  
EmployeeCount 0  
EmployeeNumber 0  
EnvironmentSatisfaction 0  
Gender 0  
HourlyRate 0  
JobInvolvement 0  
JobLevel 0  
JobRole 0  
JobSatisfaction 0  
MaritalStatus 0  
MonthlyIncome 0  
MonthlyRate 0  
NumCompaniesWorked 0  
Over18 0  
OverTime 0  
PercentSalaryHike 0  
PerformanceRating 0  
RelationshipSatisfaction 0  
StandardHours 0  
StockOptionLevel 0  
TotalWorkingYears 0  
TrainingTimesLastYear 0  
WorkLifeBalance 0  
YearsAtCompany 0  
YearsInCurrentRole 0  
YearsSinceLastPromotion 0  
YearsWithCurrManager 0  
dtype: int64
```

3.

```
Duplicates Removed: 0
```

5.

```
===== DATA TYPES BEFORE =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   1470 non-null   int64
1   Attrition                           1470 non-null   object
2   BusinessTravel                       1470 non-null   object
3   DailyRate                           1470 non-null   int64
4   Department                           1470 non-null   object
5   DistanceFromHome                     1470 non-null   int64
6   Education                           1470 non-null   int64
7   EducationField                       1470 non-null   object
8   EmployeeCount                       1470 non-null   int64
9   EmployeeNumber                      1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                              1470 non-null   object
12  HourlyRate                          1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                            1470 non-null   int64
15  JobRole                             1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                      1470 non-null   object
18  MonthlyIncome                      1470 non-null   int64
19  MonthlyRate                         1470 non-null   int64
20  NumCompaniesWorked                  1470 non-null   int64
21  Over18                             1470 non-null   object
22  OverTime                           1470 non-null   object
23  PercentSalaryHike                   1470 non-null   int64
24  PerformanceRating                   1470 non-null   int64
25  RelationshipSatisfaction             1470 non-null   int64
26  StandardHours                      1470 non-null   int64
27  StockOptionLevel                    1470 non-null   int64
28  TotalWorkingYears                   1470 non-null   int64
29  TrainingTimesLastYear               1470 non-null   int64
30  WorkLifeBalance                     1470 non-null   int64
31  YearsAtCompany                      1470 non-null   int64
32  YearsInCurrentRole                  1470 non-null   int64
33  YearsSinceLastPromotion              1470 non-null   int64
34  YearsWithCurrManager                 1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

===== DATA TYPES AFTER =====

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1470 entries, 0 to 1469

Data columns (total 35 columns):

#	Column	Non-Null Count	Dtype
0	Age	1470 non-null	int64
1	Attrition	1470 non-null	object
2	BusinessTravel	1470 non-null	object
3	DailyRate	1470 non-null	int64
4	Department	1470 non-null	object
5	DistanceFromHome	1470 non-null	int64
6	Education	1470 non-null	int64
7	EducationField	1470 non-null	object
8	EmployeeCount	1470 non-null	int64
9	EmployeeNumber	1470 non-null	int64
10	EnvironmentSatisfaction	1470 non-null	int64
11	Gender	1470 non-null	object
12	HourlyRate	1470 non-null	int64
13	JobInvolvement	1470 non-null	int64
14	JobLevel	1470 non-null	int64
15	JobRole	1470 non-null	object
16	JobSatisfaction	1470 non-null	int64
17	MaritalStatus	1470 non-null	object
18	MonthlyIncome	1470 non-null	int64
19	MonthlyRate	1470 non-null	int64
20	NumCompaniesWorked	1470 non-null	int64
21	Over18	1470 non-null	object
22	OverTime	1470 non-null	object
23	PercentSalaryHike	1470 non-null	int64
24	PerformanceRating	1470 non-null	int64
25	RelationshipSatisfaction	1470 non-null	int64
26	StandardHours	1470 non-null	int64
27	StockOptionLevel	1470 non-null	int64
28	TotalWorkingYears	1470 non-null	int64
29	TrainingTimesLastYear	1470 non-null	int64
30	WorkLifeBalance	1470 non-null	int64
31	YearsAtCompany	1470 non-null	int64
32	YearsInCurrentRole	1470 non-null	int64
33	YearsSinceLastPromotion	1470 non-null	int64
34	YearsWithCurrManager	1470 non-null	int64

dtypes: int64(26), object(9)

memory usage: 402.1+ KB

7.

The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'EX2' with files '2a.py', '2b.py', 'cleaned_employee.csv', and 'cleaned_sales_dataset.csv'. The code editor displays the contents of 'cleaned_employee.csv', which is a CSV file with columns: Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, Employee. The data is truncated in the image.

8.

```

===== CLEANING SUMMARY =====
Records Before Cleaning : 1470
Records After Cleaning  : 1356
Missing Values Found    : 0
Missing Values Removed  : 0
Duplicates Removed      : 0
Salary Outliers Removed : 114

Cleaned dataset saved as:
cleaned_employee.csv

```

PART B – DIRTY CARS DATA

Perform the following data cleaning tasks:

1. Load the dataset and display basic information.
2. Identify columns containing missing values and handle them appropriately.
3. Detect and remove duplicate transactions.
4. Find records with negative or zero Quantity and UnitPrice values and clean the dataset.
5. Convert InvoiceDate into a proper date-time format.
6. Standardize text fields such as Description by removing extra spaces and converting to a consistent case.
7. Generate a summary showing:
8. Number of missing values removed
9. Number of duplicate records removed
10. Number of invalid transactions removed
11. Save the cleaned dataset as a new CSV file.

CODE:

```
import pandas as pd
```

1. Load Dataset

```
df = pd.read_csv( r"D:\SEM_5\ML\datasets\Data cleaning\Dirty Cars\archive\dirty_cafe_sales.csv")
print("Original Shape:", df.shape)
records_before = len(df)
```

2. Basic Information

```
print("\nDataset Info")
df.info()
```

3. Missing Values

```
print("\nMissing Values")
print(df.isnull().sum())
missing_before = df.isnull().sum().sum()
for col in df.columns:
    if df[col].dtype == "object":
        df[col] = df[col].fillna(df[col].mode()[0])
missing_after = df.isnull().sum().sum()
missing_removed = missing_before - missing_after
```

4. Remove Duplicate Transactions

```
duplicates_removed = df.duplicated().sum()
df = df.drop_duplicates()
```

5. Convert Numeric Columns

```
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Price Per Unit"] = pd.to_numeric(df["Price Per Unit"], errors="coerce")
df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors="coerce")
df["Quantity"] = df["Quantity"].fillna(df["Quantity"].median())
df["Price Per Unit"] = df["Price Per Unit"].fillna( df["Price Per Unit"].median())
df["Total Spent"] = df["Total Spent"].fillna( df["Total Spent"].median())
```

6. Remove Invalid Transactions

```
invalid_transactions = len( df[ (df["Quantity"] <= 0) | (df["Price Per Unit"] <= 0)])
df = df[ (df["Quantity"] > 0) & (df["Price Per Unit"] > 0)]
```

7. Convert Transaction Date

```
df["Transaction Date"] = pd.to_datetime( df["Transaction Date"], errors="coerce")
invalid_dates = df["Transaction Date"].isnull().sum()
df = df.dropna(subset=["Transaction Date"])
```

8. Standardize Item Names

```
df["Item"] = ( df["Item"].astype(str).str.strip().str.lower())
df["Payment Method"] = (df["Payment Method"].astype(str).str.strip().str.title())
df["Location"] = (df["Location"] .astype(str).str.strip().str.title())
```

9. Save Clean Dataset

```
df.to_csv("cleaned_sales_dataset.csv", index=False)
records_after = len(df)
```

10. Cleaning Summary

```
print("Records Before Cleaning :", records_before)
print("Records After Cleaning :", records_after)
print("Missing Values Removed :", missing_removed)
print("Duplicate Records Removed :", duplicates_removed)
print("Invalid Transactions Removed :", invalid_transactions)
print("Invalid Dates Removed :", invalid_dates)
print("\nCleaned dataset saved as:")
print("cleaned_sales_dataset.csv")
```

OUTPUT:

1.

```
Original Shape: (10000, 8)
```

2.

```
Dataset Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 8 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Transaction ID         10000 non-null object
 1   Item                   9667 non-null  object
 2   Quantity               9862 non-null  object
 3   Price Per Unit         9821 non-null  object
 4   Total Spent            9827 non-null  object
 5   Payment Method         7421 non-null  object
 6   Location                6735 non-null  object
 7   Transaction Date       9841 non-null  object
dtypes: object(8)
memory usage: 625.1+ KB
```

3.

```
Missing Values
Transaction ID      0
Item                333
Quantity            138
Price Per Unit      179
Total Spent         173
Payment Method      2579
Location            3265
Transaction Date     159
dtype: int64
```

9.

```
File Edit Selection View Go Run ... EX2
EXPLORER
EX2
  .dist
  2a.py
  2b.py
  cleaned_employee.csv
  cleaned_sales_dataset.csv
cleaned_sales_dataset.csv
189 TXN_5115080,juice,5.0,3.0,15.0,Credit Card,Takeaway,2023-02-18
190 TXN_9788956,juice,3.0,3.0,12.0,Error,Takeaway,2023-04-11
191 TXN_6311602,juice,3.0,3.0,9.0,Credit Card,Takeaway,2023-05-15
192 TXN_2950680,coffee,2.0,2.0,4.0,Credit Card,Takeaway,2023-12-10
193 TXN_6749847,cake,2.0,3.0,6.0,Credit Card,Takeaway,2023-08-15
194 TXN_2143187,smoothie,1.0,4.0,4.0,Digital Wallet,In-Store,2023-04-21
195 TXN_8964522,juice,3.0,3.0,9.0,Credit Card,Takeaway,2023-10-28
196 TXN_4088917,coffee,3.0,2.0,6.0,Cash,Takeaway,2023-02-04
197 TXN_3779366,cookie,1.0,1.0,1.0,Digital Wallet,In-Store,2023-04-15
198 TXN_7724048,smoothie,1.0,4.0,4.0,Error,In-Store,2023-11-12
199 TXN_2698591,tea,5.0,1.5,7.5,Digital Wallet,Takeaway,2023-08-05
200 TXN_3171373,smoothie,3.0,4.0,12.0,Credit Card,In-Store,2023-05-10
201 TXN_6498163,smoothie,5.0,4.0,20.0,Digital Wallet,In-Store,2023-07-15
202 TXN_6717827,error,3.0,5.0,15.0,Digital Wallet,Takeaway,2023-11-15
203 TXN_9089898,salad,3.0,3.0,15.0,Credit Card,Takeaway,2023-01-11
204 TXN_8587583,cake,1.0,3.0,3.0,Digital Wallet,Takeaway,2023-10-01
205 TXN_8693704,salad,3.0,5.0,25.0,Cash,In-Store,2023-05-04
206 TXN_3857960,salad,4.0,5.0,20.0,Cash,In-Store,2023-04-26
207 TXN_3499124,error,4.0,4.0,16.0,Credit Card,Takeaway,2023-08-25
208 TXN_9250710,salad,1.0,5.0,5.0,Cash,In-Store,2023-05-28
209 TXN_7058377,juice,1.0,3.0,3.0,Digital Wallet,Takeaway,2023-04-30
210 TXN_6766492,salad,1.0,5.0,5.0,Digital Wallet,Error,2023-01-14
211 TXN_9879996,coffee,4.0,2.0,8.0,Digital Wallet,In-Store,2023-03-01
212 TXN_4291940,cake,3.0,3.0,9.0,Credit Card,In-Store,2023-11-13
213 TXN_7433461,cookie,2.0,1.0,2.0,Digital Wallet,Takeaway,2023-07-09
214 TXN_4068262,salad,2.0,5.0,10.0,Credit Card,Takeaway,2023-05-13
215 TXN_5047447,salad,1.0,5.0,5.0,Digital Wallet,Takeaway,2023-12-27
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\SEM_5\ML\LAB\EX2> python -u "d:\SEM_5\ML\LAB\EX2\2b.py"
Location 3265
Transaction Date 159
dtype: int64
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Plain Text Go Live Prettier
```

10.

```
===== CLEANING SUMMARY =====
Records Before Cleaning : 10000
Records After Cleaning : 9540
Missing Values Removed : 6826
Duplicate Records Removed : 0
Invalid Transactions Removed : 0
Invalid Dates Removed : 460

Cleaned dataset saved as:
cleaned_sales_dataset.csv
```